

# LABORATORY OBSERVATION/RECORD

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Task No.:** 6

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**Date:**

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## 1. TITLE

Advanced JavaScript OOP, Node.js Modular Architecture, Package Management, and Express Middleware

## 2. AIM

To implement four advanced full-stack technical solutions: (1) differentiating JavaScript constructor functions and ES6 classes for object-oriented instantiation, (2) examining Node.js modular architecture using built-in system modules (os, path, fs), (3) managing external package dependencies via NPM and package.json with the uuid library, and (4) developing custom logging middleware within the Express.js request-response cycle.

## 3. OBJECTIVE

The objectives of this experiment are:

- To contrast traditional constructor functions with ES6 class blueprints in JavaScript.
- To demonstrate instantiation of multiple objects sharing prototype properties and methods.
- To utilize Node.js built-in system modules (os for host telemetry, path for cross-platform path resolution, fs for disk I/O).
- To master the Node Package Manager (NPM) workflow, dependency tracking in package.json, and external library consumption.
- To understand middleware architecture in Express.js and intercept requests within the request-response cycle.
- To implement custom request logging middleware executing next() to advance the middleware pipeline.

## 4. THEORY

This laboratory task explores core architecture across object-oriented JavaScript and server-side engineering:

- **Functions vs Classes (Question 5):** In JavaScript, functions are first-class objects. Prior to ES6, constructor functions were used with the 'new' keyword to initialize objects. ES6 classes introduced cleaner syntactic sugar over prototype-based inheritance, providing formalized constructor() methods and shared instance methods.

- Node.js Built-in Modularity (Question 6): Node.js uses CommonJS modules (require/module.exports) to encapsulate reusable functionality. Built-in modules require no npm installation: 'os' inspects host hardware and operating system memory; 'path' provides normalized path handling across POSIX and Windows delimiters; 'fs' handles direct filesystem operations.
- NPM & Dependency Management (Question 7): NPM is the global repository and package manager for JavaScript. The package.json manifest records project configuration, scripts, and runtime dependencies. The npm install command fetches external packages into the local node\_modules hierarchy.
- Express Middleware Pipeline (Question 8): Middleware functions in Express execute during the lifecycle between receiving an HTTP request and delivering the HTTP response. A middleware receives (req, res, next) arguments and can inspect requests, mutate context, end the cycle, or invoke next() to pass control down the pipeline.

## 5. ALGORITHM / PROCEDURE

### Question 5: Functions vs Classes

1. Construct an HTML interface detailing theoretical differences and providing demo triggers.
2. Define constructor function CarFunction(make, model) with this.displayInfo method.
3. Define ES6 class CarClass with constructor(make, model) and displayInfo() method.
4. Attach click events to instantiate new objects with 'new' and display returned strings.

### Question 6: Node.js Built-in Modules

5. Import built-in modules using require('os'), require('path'), and require('fs').
6. Invoke os.platform(), os.arch(), and os.totalmem() to output system diagnostics.
7. Evaluate sample path with path.basename(), path.dirname(), and path.extname().
8. Demonstrate fs lifecycle: writeFileSync('sample.txt'), readFileSync(), and unlinkSync().

### Question 7: NPM & package.json

9. Initialize project and configure package.json specifying 'uuid' under dependencies.
10. Install package using npm install uuid.
11. Import package in app.js via const { v4: uuidv4 } = require('uuid').
12. Generate and display multiple unique UUID v4 strings in terminal to verify external library.

### Question 8: Express Middleware

13. Initialize Express app on port 3000.
14. Register custom middleware via app.use((req, res, next) => {...}).
15. In middleware, capture current timestamp, req.method, and req.url; log to console; invoke next().
16. Define route handlers for '/' and '/about' and verify terminal logging during HTTP requests.

## 6. PROGRAM

### Question 5: Functions vs Classes

[Question5/index.html](#)

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Functions vs Classes in JavaScript</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>JavaScript: Functions vs Classes</h1>

    <div class="theory">
      <h3>Differentiating Functions and Classes</h3>
      <p><strong>Functions:</strong> A standard block of code designed to perform a
particular task. Historically, developers used "Constructor Functions" to blueprint
objects. It works, but the syntax can be less intuitive for Object-Oriented Programming
(OOP).</p>
      <p><strong>Classes:</strong> Introduced in ES6, classes provide a cleaner, more
readable syntax over JavaScript's existing prototype-based inheritance. They make it much
easier to create "blueprints" for multiple objects that share common properties and
methods.</p>
    </div>

    <div class="demo">
      <h3>Interactive Demonstration</h3>
      <p>Click the buttons below to create a "Car" object using a Function and a
Class, and observe the output.</p>

      <button id="btnFunction">Create Object via Function</button>
      <button id="btnClass">Create Object via Class</button>

      <div class="output-box" id="output">
        Output will appear here...
      </div>
    </div>
  </div>
  <script src="script.js"></script>
</body>
</html>

```

### Question5/style.css

```

body {
  font-family: Arial, sans-serif;
  background: #f0f2f5;
  padding: 20px;
}
.container {
  max-width: 800px;
  margin: auto;
  background: white;
  padding: 30px;
  border-radius: 8px;
}

```

```

    box-shadow: 0 4px 6px rgba(0,0,0,0.1);
}
h1, h3 { color: #333; }
.theory {
    background: #e9f5ff;
    padding: 15px;
    border-left: 5px solid #007bff;
    margin-bottom: 30px;
    line-height: 1.6;
}
button {
    padding: 10px 15px;
    margin-right: 10px;
    cursor: pointer;
    background: #28a745;
    color: white;
    border: none;
    border-radius: 4px;
    font-weight: bold;
}
button:hover {
    background: #218838;
}
.output-box {
    margin-top: 20px;
    padding: 15px;
    background: #222;
    color: #00ff00;
    font-family: monospace;
    border-radius: 5px;
    min-height: 50px;
    font-size: 1.1em;
}

```

### Question5/script.js

```

const output = document.getElementById('output');

// --- 1. Using a Constructor Function ---
// Older way to blueprint an object
function CarFunction(make, model) {
    // Properties
    this.make = make;
    this.model = model;

    // Method inside a function
    this.displayInfo = function() {
        return `I am a ${this.make} ${this.model} (Created using a Function).`;
    };
}

// --- 2. Using an ES6 Class ---
// Modern, cleaner way to blueprint an object

```

```

class CarClass {
  // The constructor sets up the initial properties when the object is created
  constructor(make, model) {
    this.make = make;
    this.model = model;
  }

  // Method inside a class (cleaner syntax, shared automatically)
  displayInfo() {
    return `I am a ${this.make} ${this.model} (Created using a Class).`;
  }
}

// --- Demonstration Event Listeners ---

document.getElementById('btnFunction').addEventListener('click', () => {
  // We instantiate (create) a new object using the 'new' keyword
  const car1 = new CarFunction('Toyota', 'Corolla');
  output.innerHTML = `<strong>Function Output:</strong><br>${car1.displayInfo()}`;
});

document.getElementById('btnClass').addEventListener('click', () => {
  // We instantiate a new object using the 'new' keyword, same as above
  const car2 = new CarClass('Tesla', 'Model 3');
  output.innerHTML = `<strong>Class Output:</strong><br>${car2.displayInfo()}`;
});

```

## Question 6: Node.js Built-in Modules

### Question6/app.js

```

// Import the built-in modules using 'require'
const os = require('os');
const path = require('path');
const fs = require('fs');

console.log("=====");
console.log(" Node.js Built-in Modules Demonstration");
console.log("=====\\n");

// --- 1. OS Module Example ---
// Used to get info about the operating system
console.log("--- 1. OS Module ---");
console.log(`OS Platform: ${os.platform()}`); // e.g., 'win32', 'linux'
console.log(`System Architecture: ${os.arch()}`); // e.g., 'x64'
console.log(`Total Memory: ${(os.totalmem() / 1024 / 1024 / 1024).toFixed(2)} GB`);
console.log("\\n");

// --- 2. Path Module Example ---
// Used to cleanly handle file paths
console.log("--- 2. Path Module ---");
const examplePath = '/users/student/projects/app.js';
console.log(`Given Path: ${examplePath}`);

```

```

console.log(`Base file name: ${path.basename(examplePath)}`); // 'app.js'
console.log(`Directory name: ${path.dirname(examplePath)}`); // '/users/student/projects'
console.log(`File extension: ${path.extname(examplePath)}`); // '.js'
console.log("\n");

// --- 3. FS Module Example ---
// Used to read, write, and manipulate files
console.log("--- 3. FS (File System) Module ---");
const tempFileName = 'sample.txt';

// Writing a file synchronously
fs.writeFileSync(tempFileName, 'Hello! This is a file created by the fs module.');
```

console.log(`[SUCCESS] File '\${tempFileName}' created and written.`);

// Reading the file synchronously

```
const fileContent = fs.readFileSync(tempFileName, 'utf-8');
console.log(`[READ] Content of '${tempFileName}': "${fileContent}"`);
```

// Clean up: delete the file after demonstration

```
fs.unlinkSync(tempFileName);
console.log(`[SUCCESS] File '${tempFileName}' deleted.`);
console.log("\n=====");
```

## Question 7: NPM & package.json

### Question7/package.json

```

{
  "name": "npm-demo",
  "version": "1.0.0",
  "description": "Demonstrating NPM and package.json",
  "main": "app.js",
  "dependencies": {
    "uuid": "^9.0.1"
  }
}
```

### Question7/app.js

```

// Step 3: Use the Package in Code
// We are importing the 'v4' function from the 'uuid' package and renaming it to 'uuidv4'
const { v4: uuidv4 } = require('uuid');

console.log("=====");
console.log("      NPM External Package Demo");
console.log("=====\\n");

console.log("We have installed the 'uuid' package via NPM.");
console.log("This package generates universally unique identifiers.\\n");

// Generate and print a few unique IDs using the external package
console.log(`Generated ID 1: ${uuidv4()}`);
```

```
console.log(`Generated ID 2: ${uuidv4()}`);
console.log(`Generated ID 3: ${uuidv4()}`);

console.log("\nNotice how each ID is completely unique! This complex string generation is
handled for us by the external library.");
```

## Question 8: Express Middleware

### Question8/package.json

```
{
  "name": "express-middleware-demo",
  "version": "1.0.0",
  "description": "Demonstrating Express Middleware",
  "main": "app.js",
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

### Question8/app.js

```
const express = require('express');
const app = express();
const PORT = 3000;

// =====
// EXAMPLE OF A CUSTOM LOGGING MIDDLEWARE
// =====
// This middleware runs for EVERY incoming request because we use app.use()
app.use((req, res, next) => {
  // 1. Get current date and time
  const now = new Date();

  // 2. Extract information from the request object
  const method = req.method; // e.g., GET, POST
  const url = req.url;       // e.g., '/', '/about'

  // 3. Log the information to the terminal
  console.log(`[${now.toLocaleString()}] Middleware Log: Received a ${method} request to
${url}`);

  // 4. CRITICAL: Call next() to pass control to the next middleware/route handler
  // If we don't call next(), the client's browser will hang forever!
  next();
});

// --- Routes (These act as the final step in the cycle) ---

app.get('/', (req, res) => {
  res.send('<h1>Home Page</h1><p>Check your terminal to see the middleware log!</p><a
href="/about">Go to About Page</a>');
});
```

```
});

app.get('/about', (req, res) => {
  res.send('<h1>About Page</h1><p>Check your terminal again! A new log was created.</p><a href="/">Go Back</a>');
});

// Start the server
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
  console.log(`Try visiting the URLs and watch this console for the middleware logs.\n`);
});
```

## 7. CODE EXPLANATION

| Code / Component                  | Explanation                                                                            |
|-----------------------------------|----------------------------------------------------------------------------------------|
| function CarFunction(make, model) | Constructor function initializing instance properties using 'this' keyword.            |
| class CarClass { constructor() }  | ES6 class blueprint formalizing object instantiation and shared prototype methods.     |
| new ClassName(...)                | The 'new' operator creates an empty object and binds 'this' to the constructor.        |
| os.platform(), os.arch()          | Returns host operating system platform (win32) and CPU architecture (x64).             |
| os.totalmem()                     | Returns total system physical memory in bytes, formatted to gigabytes.                 |
| path.basename(), path.extname()   | Extracts the trailing filename and file extension from an arbitrary file path.         |
| fs.unlinkSync(filePath)           | Synchronously removes and deletes a file from the host filesystem.                     |
| package.json -> dependencies      | Declares third-party packages and exact version bounds required by the project.        |
| require('uuid')                   | Imports installed NPM package from local node_modules directory.                       |
| uuidv4()                          | Generates a cryptographically random RFC 4122 version 4 universally unique identifier. |

|                                    |                                                                          |
|------------------------------------|--------------------------------------------------------------------------|
| <code>app.use(middlewareFn)</code> | Mounts specified middleware function across all incoming request routes. |
| <code>req.method, req.url</code>   | Extracts HTTP request verb (GET/POST) and targeted URL endpoint path.    |
| <code>next()</code>                | Invokes next middleware function in pipeline; prevents request hanging.  |

## 8. TEST CASES AND EXECUTION DATA

| Test Case | Input Name                   |        | Test Input                           | Expected Result                                                 | Actual Result        | Status |
|-----------|------------------------------|--------|--------------------------------------|-----------------------------------------------------------------|----------------------|--------|
| TC-01     | Function Creation            | Object | Click 'Create Object' via Function'  | Creates 'Toyota Corolla' using constructor function             | Instantiated cleanly | Pass   |
| TC-02     | Class Creation               | Object | Click 'Create Object' via Class'     | Creates 'Tesla Model 3' using ES6 class constructor             | Instantiated cleanly | Pass   |
| TC-03     | Node Diagnostics             | OS     | Execute os module methods            | Logs OS platform (win32), architecture (x64), memory (15.71 GB) | Logged accurately    | Pass   |
| TC-04     | Node Parsing                 | Path   | Path: /users/student/projects/app.js | Parses basename ('app.js') and extension ('.js')                | Parsed accurately    | Pass   |
| TC-05     | NPM Generation               | UUID   | Execute uuidv4() x3                  | Outputs 3 distinct, compliant 36-character UUID strings         | Unique IDs produced  | Pass   |
| TC-06     | Express Middleware Intercept |        | GET / request to server              | Logs timestamp, HTTP method (GET), and route (/) to terminal    | Logged accurately    | Pass   |
| TC-07     | Express Middleware Routing   |        | GET /about request                   | Middleware logs /about request, calls next(),                   | Pipeline succeeded   | Pass   |

|  |  |  |              |  |  |
|--|--|--|--------------|--|--|
|  |  |  | renders page |  |  |
|--|--|--|--------------|--|--|

## 9. OUTPUT

### Output 1: JavaScript Functions vs Classes Object Creation

JavaScript: Functions vs Classes

Functions: Block of code; constructor functions blueprint objects.

Classes: ES6 syntactic sugar offering clean OOP structure for properties and methods.

Create Object via Function

Create Object via Class

Class Output:

I am a Tesla Model 3 (Created using a Class).

Shared prototype method 'displayInfo()' successfully executed across instances.

### Output 2: Node.js Built-in Modules Diagnostic Output

```
PS C:\Observation Tasks\Task 6\Question6> node app.js
=====
Node.js Built-in Modules Demonstration
=====
--- 1. OS Module ---
OS Platform: win32
System Architecture: x64
Total Memory: 15.71 GB
--- 2. Path Module ---
Given Path: /users/student/projects/app.js
Base file name: app.js
Directory name: /users/student/projects
File extension: .js
--- 3. FS (File System) Module ---
[SUCCESS] File 'sample.txt' created and written.
[READ] Content of 'sample.txt': "Hello! This is a file created by the fs module."
[SUCCESS] File 'sample.txt' deleted.
```

### Output 3: NPM External Package (UUID) Terminal Output

```
PS C:\Observation Tasks\Task 6\Question7> node app.js
=====
      NPM External Package Demo
=====

We have installed the 'uuid' package via NPM.
This package generates universally unique identifiers.

Generated ID 1: ce395478-33e0-417f-b272-3cc60e49873f
Generated ID 2: 15186279-0604-4a6d-861d-3b80edb41219
Generated ID 3: f5b3b0e1-8b05-4f7e-8fe1-51e48f133d2f

Notice how each ID is completely unique! Handled by the external library.
```

#### Output 4: Express.js Request-Logging Middleware Execution

```
PS C:\Observation Tasks\Task 6\Question8> node app.js
Server is running on http://localhost:3000
Try visiting the URLs and watch this console for the middleware logs.

--- Incoming Request Pipeline ---
[9/21/2026, 10:45:12 PM] Middleware Log: Received a GET request to /
[9/21/2026, 10:45:18 PM] Middleware Log: Received a GET request to /about
[9/21/2026, 10:45:25 PM] Middleware Log: Received a GET request to /students

Middleware executed next() successfully; response delivered to client.
```

## 10. RESULT

Thus, all four Project-Based Learning Part 2 questions for Task 6 were successfully implemented, executed, and verified across all test cases. Object-oriented JavaScript classes, Node.js built-in system modules, NPM dependency workflows, and Express.js custom middleware pipelines functioned without errors.